

ibm370_channel_sop_v1json

```

{
  "id": "SOP-IBM370-002",
  "title": "IBM System/370 Channel Check & I/O Error Isolation",
  "system": "IBM System/370",
  "component": "I/O Channels (Selector / Multiplexer)",
  "author": "SymbioStreams Semantic Bridge",
  "source": "IBM Field Engineering Maintenance Manual (SY24-3581)",
  "description": "Diagnostic procedure for isolating channel-level errors and interface parity
checks on the Model 145/155.",
  "steps": [
    {
      "step": 1,
      "action": "Channel Status Word (CSW) Retrieval",
      "details": "When an I/O interruption occurs, examine the CSW at real address 64.
Identify if bit 44 (Channel Data Check) or bit 45 (Channel Control Check) is set.",
      "check": "Bit 45 indicates an internal channel hardware failure."
    },
    {
      "step": 2,
      "action": "Limited Channel Log (LCL) Analysis",
      "details": "Read the Limited Channel Log stored in storage locations 176-179. This
contain the logout data at the time of the channel check.",
      "indicator": "Identify the failing subchannel address and the specific error type
(Interface Control Check vs. Internal Check)."
    },
    {
      "step": 3,
      "action": "Bus In/Out Parity Verification",
      "details": "Check the channel console lights for 'Bus In' or 'Bus Out' parity errors.
This often indicates a loose cable or a failing tag-line driver on the I/O interface.",
      "logic": "If Bus Out parity is on, the error is likely in the CPU/Channel logic. If Bus
In is on, the error is in the device control unit."
    },
    {
      "step": 4,
      "action": "Run Micro-Diagnostic Loops",
      "details": "Load the Channel Micro-Diagnostic routines from the console file. Set the
loops to repeat on the failing subchannel.",
      "remedy": "Isolate the failure to the specific Channel Logic board (MST) and replace."
    }
  ],
  "diagnostic_flowchart": [
    {
      "condition": "Channel Control Check ON",
      "next": "Internal channel failure. Run micro-diagnostics."
    },
    {
      "condition": "Interface Control Check ON",
      "next": "Check control unit cables and power."
    },
    {
      "condition": "Bus In Parity error",
      "next": "Isolate to device control unit or interface terminator."
    }
  ]
}

```